SENG 430 — Software Quality Assurance Laboratory

HealthML

ML Clinical Visualizer for Healthcare Professionals

7 Pipeline Steps | 20 Clinical Domains | 6 ML Models | 3 Docker Services

Group: KaraKartallar

Instructor: Dr. Sevgi KOYUNCU TUNÇ
Assistant: Dr. Buse EROL ESİRİK

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👋 Team KaraKartallar

Roles & Responsibilities

NAME	ROLE
Mustafa Faruk Ekşi	Product Owner
Şahin Erşan	QA / Documentation Lead, Developer
Özgün Soykök	UX Designer, Developer
Ceren Tekin	Lead Developer
İsmail Doğan	Scrum Master

📋 What We Built

An end-to-end clinical ML pipeline visualizer that teaches healthcare professionals how machine learning works — from data to diagnosis, with real models, real metrics, and real fairness auditing.

🎯 Target Users

Medical students, clinicians, and health informatics professionals who need to understand ML without writing code.

❓ The Problem

Healthcare professionals need to understand ML decisions, but existing tools are too technical. HealthML bridges this gap with a guided, no-code clinical pipeline.

🏗️ System Architecture

Three-Tier Microservice Design

Frontend

React 19 · Vite 8
TailwindCSS 4 · shadcn/ui
Recharts · Framer Motion

Port :80 (Nginx)

→

Backend API

Node.js · Express 5
TypeScript · Multer
PapaParse · Proxy Layer

Port :3001

→

ML Service

Python 3.11 · FastAPI
scikit-learn · SMOTE
fpdf2 · pandas

Port :8000

🐳 Docker Deployment

3 isolated containers orchestrated with Docker Compose. Deployed on GCP Compute Engine. Nginx reverse-proxies /api/* to the backend.

🔄 Data Flow

Frontend manages state via React Context. Backend proxies ML operations to Python. Models stored in-memory keyed by UUID. PDF certificates generated server-side.

🔬 7-Step Clinical ML Pipeline

Guided workflow enforcing correct learning sequence

1. Clinical Context — Choose specialty
2. Data Exploration — Upload & map CSV
3. Data Preparation — Clean, normalize, SMOTE
4. Model & Parameters — Pick algorithm + tune
5. Results — Metrics, confusion matrix, ROC
6. Explainability — Feature importance + patient waterfall
7. Ethics & Bias — Fairness audit + PDF certificate

🔒 Step Access Control

Users cannot skip steps. Schema must be validated before Step 3. Data must be prepared before Step 4. This mirrors real clinical ML workflow constraints.

📖 ML Glossary

20 key ML terms explained in plain language accessible via a Help button. Designed for clinicians who are not ML experts.

🖥 Live Demo

Full Steps 1–7 Walkthrough

Domain 1: Cardiology | **Domain 2: Oncology — Breast** | **Domain 3: Endocrinology — Diabetes**

🌐 Switch to: **http://www.healthml.com.tr/**

Show 3 domains · Train models · Compare results · Show explainability · Show bias table

⏱️ Budget: 4 minutes for live demo (1:00 – 5:00)

📊 Results Interpretation

What do these metrics mean for real patients?

Sensitivity (Recall) — "catch rate"
Of all patients who truly have the disease, what percentage did our model correctly identify?

Example: A Sensitivity of 85% means we catch 85 out of 100 truly sick patients. The remaining 15 are *missed* — false negatives.

Clinical Impact of Low Sensitivity
In healthcare, missing a sick patient (false negative) can be **life-threatening**. A model with 60% sensitivity misses 4 in 10 patients — unacceptable for screening tools.

This is why we prioritize Sensitivity over Accuracy in clinical ML.

🎯 Metric Thresholds (Built-in)

METRIC	GREEN	AMBER
Accuracy	≥80%	≥70%
Sensitivity	≥75%	≥50%
Specificity	≥80%	≥70%
AUC	≥0.85	≥0.70

🖥️ Show in live app: Train Random Forest on Cardiology domain → point to color-coded metric cards

⚖️ Bias & Fairness Audit

Subgroup analysis — gender & age bands

How It Works

- Model is evaluated separately for each subgroup (Male/Female, Age bands)
- Sensitivity per subgroup is compared to overall sensitivity
- **Alert triggered** when any gap exceeds **10 percentage points**
- Age bands: raw values use fixed ranges; normalized values use quantile-based tertiles

⚠️ **Example disparity:** If overall sensitivity = 82% but Female subgroup = 68%, the gap is 14 pp → BIAS ALERT

EU AI Act Compliance Checklist

- ✅ Human oversight mechanism
- ✅ Transparency in decision-making
- ✅ Data quality documentation
- ✅ Bias monitoring & mitigation
- ✅ Purpose limitation statement
- ✅ Risk classification acknowledgment
- ✅ Audit trail (PDF certificate)
- ✅ Educational disclaimer

🖥️ Show Step 7 in live app → subgroup table → bias alerts → PDF certificate download

🧠 Key Technical Decisions

🔬

Real scikit-learn

Actual model training — not simulated. Metrics vary with data and hyperparameters.

🐳

3-Container Split

Frontend, Backend API, and ML Service run independently. Clean separation of concerns.

🔍

Ablation Explainability

Per-patient waterfall — works across all 6 models without SHAP/LIME dependency.

⚖️

SMOTE Train-Only

Synthetic rows tagged with sentinel column, excluded from test split. No data leakage.

📏

10pp Bias Threshold

Calibrated to avoid noise from small datasets while catching clinically meaningful gaps.

🔒

Step Gating

Enforced prerequisite completion mirrors real clinical ML workflow constraints.

📚 What We Learned

Technical lessons from building HealthML

🧪 Test early with real data

SMOTE leakage and Unicode PDF bugs were invisible with synthetic data. Real CSV uploads exposed them immediately.

📝 Plain language matters

Clinicians ignored metrics until labels were rephrased. "% of truly positive patients caught" works better than "Recall".

🏗️ Microservices add complexity

Cross-container networking and environment variables cost significant debugging time. Worth it for clean architecture, but plan for the overhead.

🚪 Gate reviews prevent debt

Enforcing sprint gate criteria (live demo + passing tests) prevented unfinished work from accumulating.

⚖️ Fairness isn't an afterthought

Retrofitting bias analysis in Sprint 4 was expensive. Building fairness hooks from Sprint 2 would have been significantly cheaper.

📈 Velocity improved over time

Sprint 1: 3 story points → Sprint 5: 11 story points. Weekly instructor feedback accelerated decisions and reduced rework.

🏆 Quality & Delivery Evidence

Testing, QA reports, and performance metrics

🧪 Testing & QA

- **Playwright E2E:** Full 7-step pipeline automated test suite
- **Sprint 4 Acceptance:** Bias threshold, checklist progress, certificate generation
- **User Testing:** 5 healthcare students, signed consent forms, pass/fail per task
- **QA Reports:** Sprint 3, 4, 5 review PDFs with corrective actions logged

⚡ Lighthouse Scores

METRIC	SCORE
Performance	>80
Accessibility	>90
Best Practices	>90
SEO	>90

📈 Sprint Velocity

Sprint 1: **3 pts** → Sprint 5: **11 pts**
Velocity increased 3.6× across 5 sprints

📦 Delivery Artifacts

- 5 Sprint Reports (PDF)
- Architecture Diagram (draw.io)
- User Testing Report
- Signed Consent Forms
- 2-Page Project Report (PDF)

Thank You!

We're ready for your questions.

🌐 **Live App**
healthml.com.tr

📂 **GitHub**
MuffinMufido/HealthML

📋 **Jira**
HCML Board

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